

E

Earth

age of 93, 115, 116
 circumference of 21, 34
 density/weight of 94, 96–97
 flat Earth theory 16, 17
 floating disk theory 11, 12
 geocentric universe 33, 46–48, 52
 geological structure 115
 rotation of 282
 solid core 217
 tectonic plates 206–207
 earthquakes 22, 217
 Edison, Thomas 159
 Edwards, Robert **261**
 Einstein, Albert 151, 168, 171, 196,
 197, **198–203**, 216, 221, 240,
 279, 299, 300
 elasticity 70, 72
 electric current 124, 139
 electrical engineering 246
 electricity
 du Fay 84
 Faraday 123–125
 Franklin 83–85
 Henry 124
 Hertz 150
 Maxwell 148–151
 Ørsted 160
 Volta 105
 electromagnetic generation 105
 electromagnetic radiation 168, 170
 electromagnetic spectrum 223
 electromagnetic transmitters 159
 electromagnetism 123, 124,
 148–151
 electron microscopes 167
 electronics 246–249
 electrons 186, 189, 196,
 234, 240, 246, 249,
 251, 280
 electrophorus 105
 elements
 classification of 153–155
 forms of 101
 four 16
 isotopes 239, 240
 modern definition of 68
 nature of 111
 radioactive 183
 elephants 116–117
 elliptical orbits 58–59
 email 307
 empirical science 11
 energy

of chemical compounds 232
 conservation of 139
 and mass 197, 201
 of molecules 234
 nuclear 199, 201
 quanta 168, 170–171, 189
 quantum mechanics 300
 transfer of 139
 engineering
 Bell 159
 Brill 310
 Bush 235
 da Vinci 43, 45
 Tereshkova 296
 Watt 97
 Enigma machines 256–257
 ENQUIRE program 305
 enzymology 236
 epicycles 48
 Eratosthenes of Cyrene 19, **21**
 ethology 231, 294
 Euclid 30, 33, 36, **38**
Eureka 19, 20
 europium 239
 event horizons 300
 evolution
 Darwin 132–137
 de Vries 142
 Dobzhansky 260
 rejected by Cuvier and Agassiz
 117
 Wallace 135
 Weismann 161
 experimental physics 252
 experimental science 67, 68
 extinctions 115–116, 252
 eyes 166, 297

F

Fabricius, Hieronymus 61
 Falloppio, Gabriele **62**
 Faraday, Michael **122–125**, 150
 Fermat, Pierre 32
 fermentation 144
 Fermi, Enrico **196**, 240
 fermions 281
 Ferranti Mark I 258
 Feynman, Richard **278–281**
 Feynman diagrams 280, 281
 Fibonacci, Leonardo **44**, 45
 Fibonacci sequence 44
 Finlay, Carlos Juan **161**
 Fisher, Ronald **219**
 fixed-action patterns 231

Fleming, Alexander **212–215**
 flight
 da Vinci 43
 Florey, Howard **215**
 Flow-Matic 242
 fluid dynamics 282
 fossils 114, 115–117, 205, 206
 Fourier, Joseph **119**
 fractal geometry 285
 François I of France 43, 45
 Franklin, Benjamin **82–85**
 Franklin, Rosalind 267–268,
270–273
 frequency 150
 friction 105
 Frisch, Otto 196–197
 fruit flies 180–181

G

Galápagos Islands 133, 134
 galaxies 221–223
 Galen, Claudius **24–29**, 61
 Galileo Galilei 16–17, 48, **50–55**,
 67, 68
 gallium 154
 Galvani, Luigi 105
 gamma radiation 186
 gases
 in air 101, 102
 atmospheric 111–112
 Boyle's law 67, 68, 69, 72
 expansion and temperature
 changes 139
 Gay-Lussac's Law 68
 generating and separating 94–96
 kinetic theory 150
 liquefaction 124
 Gassendi, Pierre **63**
 Gay-Lussac, Joseph Louis **68**
 Ge Hong 292–293
 Geiger counters 253
 Gell-Mann, Murray **281**
 gene therapy 274
 generators, electric 124, 125
 genes 140, 142
 genetic code 236, 276
 genetic manipulation 274
 genetics
 Crick 264–269
 de Vries 142
 Dobzhansky 260
 Fisher 209
 Mak 303
 McClintock 229

Mendel 140–143
 Morgan 178–181
 Muller 180
 Ochoa 236
 Sanger 274–277
 Stevens 174
 Watson 264–269
 geocentric universe 33, 46–48, 52
 geography
 Eratosthenes 21
 Humboldt 160
 Ptolemy 33
 geology
 Agassiz 117
 Hutton 93
 Steno 118
 Tharp 207
 Wegener 205–207
 geometry 11, 12–13, 19, 30, 44, 285
 germ theory 146–147
 germanium 154
 Gilbert, Walter 274
 Gilbert, William **62–63**
 glaciers 117
 Goeppert Mayer, Maria **238–241**
 Golgi, Camillo 164, 166, **167**
 Goodall, Jane **294–295**
 gravity 58, 72, 77, 78, 79–81,
 96–97, 199, 201, 299
 Gray, Elisha 159
 Greider, Carol **311**
 Gulf Stream 83, 85, 282
 gunpowder 39

H

Hahn, Otto 196, 197, 234
 half-life 189
 Halley, Edmund 76
 Harvey, William **60–61**
 Hawking, Stephen 151, **298–301**
 heart 26, 61
 heat 103, 139, 168, 194
 Heatley, Norman 215
 Heisenberg, Werner **260**
 heliocentric universe 46–49, 51,
 57, 58, 77–78
 helium 280
 hematology 286
 Henry, Joseph **124**
 heredity, laws of 140–143,
 178–181, 266
 heresy 48, 51, 53
 Herodotus 12, 13
 Herschel, John **129**